20. A generalized version of the estimator for β2 proposed in Exercise 2.9 by Professor I.M. Mean for the regression model $\boldsymbol{y_i = \beta_1 + \beta_2 x_i + e_i, i = 1, 2, \ldots, N}$ is

$$\hat{\boldsymbol{\beta}}_{2,mean} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1}$$

Where $(\bar{y}_1, \bar{x}_1)$ and $(\bar{y}_2, \bar{x}_2)$ are the sample means for the first and second halves of the sample observations, respectively, after ordering the observations according to increasing values of x. Given that assumptions MR1–MR6 hold:

a. Show that $\hat{\boldsymbol{\beta}}_{2,mean}$ is unbiased.

Substituting the regression model into the estimator, we have

$$\bar{y}_1 = \beta_1 + \beta_2\bar{x}_1 + \bar{e}_1, \qquad \bar{y}_2 = \beta_1 + \beta_2\bar{x}_2 + \bar{e}_2$$

Thus,

$$\bar{y}_2 - \bar{y}_1 = \beta_2(\bar{x}_2 - \bar{x}_1) + (\bar{e}_2 - \bar{e}_1)$$

Substituting into the estimator gives

$$\hat{\beta}_{2,mean} = \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}$$

Taking expectations conditional on x, and using $E(e_i \mid x) = 0$, we obtain

$$E(\hat{\beta}_{2,mean}) = \beta_2$$

Therefore, $\hat{\beta}_{2,mean}$ is an unbiased estimator of β_2.

b. Derive an expression for $\boldsymbol{var(\hat{\beta}_{2,mean}|x)}$.

From part (a),

$$\hat{\beta}_{2,mean} = \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}$$

Thus,

$$Var(\hat{\beta}_{2,mean} \mid x) = \frac{Var(\bar{e}_2 - \bar{e}_1)}{(\bar{x}_2 - \bar{x}_1)^2}$$

Since the two halves of the sample are independent,

$$Var(\bar{e}_2 - \bar{e}_1) = Var(\bar{e}_2) + Var(\bar{e}_1)$$

Each sample mean is based on $N/2$ observations, so

$$Var(\bar{e}_1) = Var(\bar{e}_2) = \frac{\sigma^2}{N/2} = \frac{2\sigma^2}{N}$$

Therefore,

$$Var(\bar{e}_2 - \bar{e}_1) = \frac{2\sigma^2}{N} + \frac{2\sigma^2}{N} = \frac{4\sigma^2}{N}$$

and

$$Var(\hat{\beta}_{2,mean} \mid x) = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}$$

c. Write down an expression for $var(\widehat{\beta}_{2,mean})$.

The unconditional variance of the mean-based estimator $\hat{\beta}_{2,\text{mean}}$ is

$$\operatorname{Var}(\hat{\beta}_{2,\text{mean}}) = \frac{\operatorname{Var}(\bar{e}_2 - \bar{e}_1)}{(\bar{x}_2 - \bar{x}_1)^2} = \frac{\operatorname{Var}(\bar{e}_2) + \operatorname{Var}(\bar{e}_1)}{(\bar{x}_2 - \bar{x}_1)^2} = \frac{4\sigma^2/N}{(\bar{x}_2 - \bar{x}_1)^2}.$$

Here, we use $\operatorname{Var}(\bar{e}_1) = \operatorname{Var}(\bar{e}_2) = 2\sigma^2/N$ because each half-sample contains $N/2$observations.

Since x is fixed, the unconditional variance is the same as the conditional variance formula, depending only on the error terms.

d. Under what conditions will $\widehat{\beta}_{2,mean}$ be a consistent estimator for β_2?

$\widehat{\beta}_{2,mean}$ is a consistent estimator for β_2 if the sample size N goes to infinity and the usual linear regression assumptions (MR1–MR6) hold. These assumptions ensure that the errors have zero mean, finite variance, are uncorrelated, and that the regressors are non-stochastic or satisfy relevant conditions, so that the estimator converges in probability to the true β_2.

e. Randomly generate observations on x from a uniform distribution on the interval (0,10) for sample sizes $N = 100, 500, 1000$, and, if your software permits, $N = 5000$. Assuming $\sigma^2 = 1000$, for each sample size, compute:

i. **$var(b_2|x)$ and $var(\widehat{\beta}_{2,mean}|x)$) where b_2 is the OLS estimator.**

ii. **Estimates for $E[(s_{-1}^2)^{-1}]$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2$ where s_x^2 is the sample standard deviation for x using N as a divisor.**

N	$var(b_2\|x)$	$var(\widehat{\beta}_{2,mean}\|x)$	$1/s_x^2$	$4/(\bar{x}_2 - \bar{x}_1)^2$
100	1.2436	1.6632	0.1244	0.1663
500	0.2358	0.3118	0.1179	0.1559
1000	0.1189	0.1580	0.1189	0.1579
5000	0.0247	0.0333	0.1235	0.1666

f. Comment on the relative magnitudes of your answers in part (e), (i) and (ii) and how they change as sample size increases. Does it appear that $\widehat{\beta}_{2,mean}$ is consistent?

As the sample size increases, both $\operatorname{Var}(b_2 \mid x)$vand $\operatorname{Var}(\hat{\beta}_{2,\text{mean}} \mid x)$ decrease, indicating that both estimators become more precise with larger samples. However, $\operatorname{Var}(b_2 \mid x)$ is consistently smaller than $\operatorname{Var}(\hat{\beta}_{2,\text{mean}} \mid x)$, showing that the OLS estimator is more efficient.

Similarly, the values of $E[1/s_x^2]$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2]$ remain relatively stable as the sample size increases, but the latter is generally larger, which explains why the variance of the mean-based estimator is higher.

Since $\operatorname{Var}(\hat{\beta}_{2,\text{mean}} \mid x) \to 0$ as $N \to \infty$, it appears that $\hat{\beta}_{2,\text{mean}}$ is a consistent estimator for β_2.

g. Show that $E\left[(s_{-1}^{2})^{-1}\right] \xrightarrow{p} 0.12$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2 \xrightarrow{p} 0.16$. [Hint: The variance of a uniform random variable defined on the interval (a, b) is $(b - a)^2 / 12$.]

As the sample size increases, s_x^2 converges in probability to the population variance of a uniform distribution on (0,10), which is $(10 - 0)^2/12 = 8.3333$. Therefore, $(s_x^2)^{-1}$ converges to $1/8.3333 =$ 0.12.

For the second expression, when the sample is ordered, the first half converges to a uniform distribution on (0,5) with mean 2.5, and the second half converges to a uniform distribution on (5,10) with mean 7.5. Hence, $\bar{x}_2 - \bar{x}_1 \rightarrow 5$, and

$$\frac{4}{(\bar{x}_2 - \bar{x}_1)^2} \rightarrow \frac{4}{5^2} = 0.16.$$

h. Suppose that the observations on x were not ordered according to increasing magnitude but were randomly assigned to any position. Would the estimator $\widehat{\beta}_{2,mean}$ be consistent? Why or why not?

If the observations on x are randomly assigned rather than ordered, both subsamples will have similar distributions, so that $\bar{x}_1 \approx \bar{x}_2$. As a result, $\bar{x}_2 - \bar{x}_1 \rightarrow 0$, making the estimator unstable. Therefore, $\hat{\beta}_{2,\text{mean}}$ does not converge to β_2 and is not consistent.